

No. of Functional Features Included in the Solution

Date	3 July 2026
Team ID	XXXXXX
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Data Loading	Loads the historical flood/weather dataset for analysis and model training.	Data Collection	Done	High
2	Data Preprocessing	Handles missing values, categorical values, outliers, feature scaling, and train-test splitting.	Data Preprocessing	Done	High
3	Machine Learning Models	Implements Decision Tree, Random Forest, KNN, and XGBoost algorithms.	Model Building	Done	High
4	Model Evaluation	Compares model performance and selects the best-performing model.	Model Evaluation	Done	High
5	Flood Prediction	Predicts whether flood conditions are likely based on user input.	Prediction Engine	Done	High
6	Flask Web Application	Provides a web interface for entering weather parameters and viewing predictions.	Application Layer	Done	Medium
7	Result Display	Displays Flood Detected or No Flood Detected with appropriate result page.	Frontend	Done	Medium
8	Model Storage	Saves and loads the trained model (.pkl) for future predictions.	Model Storage	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Home page, Input Form, Prediction Result Pages
2	Backend / Logic	4	Data Preprocessing, Model Training, Prediction, Model Evaluation
3	Database / Storage	1	Dataset (CSV), Saved Model (.pkl)
4	API / Integration	1	Flask Routes connecting frontend and prediction model
5	Security / Authentication	0	Not implemented (Not required for this project)